Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Senior Full Stack Engineer with 8 years of professional experience building high-impact web applications using **TypeScript**, **React**, and **Node.js** end-to-end across B2B SaaS, healthcare, fintech, and civic tech environments. Proven track record of owning features from database to UI, designing **REST APIs**, managing **relational databases** (PostgreSQL), and deploying to **AWS** with CI/CD pipelines and Docker. Thrives in small, high-ownership teams with a collaborative engineering culture built on code reviews, mentorship, and iterative delivery.

CORE COMPETENCIES

- **Full Stack TypeScript:** End-to-end TypeScript development across React frontends and Node.js backends, with strong JavaScript (ESNext) foundations.
- **Backend & APIs:** NestJS, Node.js REST APIs, PostgreSQL, MongoDB, Prisma ORM, JWT authentication, role-based access control, and SSE streaming.
- **Frontend Frameworks:** React, Next.js, Vue.js, Redux, Zustand, React Query, Tailwind CSS, Radix UI, and Storybook.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, Docker Compose, GitHub Actions, CI/CD pipelines, Heroku, and Nx monorepo orchestration.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **Performance & Observability:** Lighthouse, Chrome DevTools, code splitting, CDN strategies, Sentry, bundle analysis, and Core Web Vitals.
- **Leadership:** Technical architecture reviews, mentorship, pair programming, code reviews, Agile/Scrum, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing adoption.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, delivering end-to-end **TypeScript** development from API layer to conversational UI.
- Implemented **JWT-based authentication** and **role-based access control** in **Next.js** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through pair programming and code reviews.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating **CI/CD** processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for real-time data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **Node.js RESTful APIs** to power new features, focusing on testable, maintainable, and well-documented code, and translated business requirements into efficient **MongoDB** database models and schemas.
- Transformed Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface aligned with product requirements.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria

Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **Node.js**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **Node.js** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications** in **TypeScript**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and built **NestJS** REST APIs powering user management, analytics, ingestion tracking, and conversational AI endpoints.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring 22 interactive chart types, dark mode, URL state management, and responsive mobile layouts.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests (4 projects), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown.
- Implemented **SSE streaming** for real-time chat responses with keepalive heartbeats, phone OTP authentication, **Telegram bot** with webhook handling, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*